Lesson 8.3 - Law of Large Numbers

> **Video:** Law of Large Numbers by The Organic Chemistry Tutor **Chapter:** 8 - Normal Probability Distributions

Statement

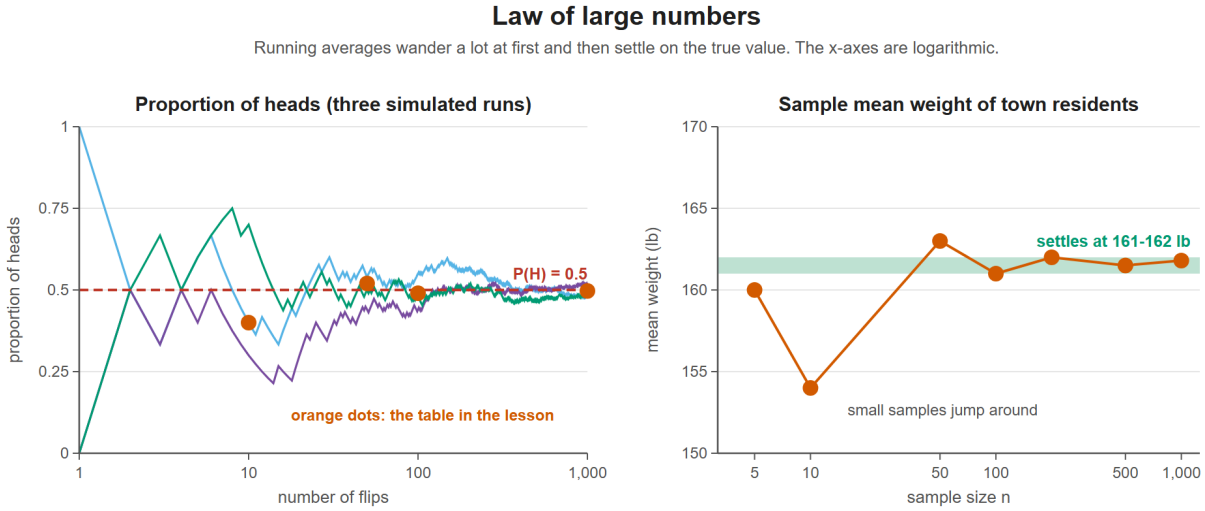
> If an experiment is repeated many times, the **observed average** (or observed proportion) gets **closer and closer to the true, theoretical value** as the number of trials increases.

Example 1 - flipping coins

The theoretical probability of heads is $P(H) = \frac{1}{2} = 0.5$. A few trials can stray far from 0.5, but many trials land close to it:

Number of flips	Heads observed (example)	Observed proportion
10	4	0.40
50	26	0.52
100	49	0.49
1,000	497	0.497

As the number of flips grows, the observed proportion **approaches 0.5**.



Left: three simulated runs of 1,000 coin flips. Each wanders early on, then all three close in on 0.5. Right: the sample mean weights from the lesson, which jump around for small samples and settle near 161-162 lb.

Example 2 - estimating an average weight

A town has 20,000 residents. Instead of weighing everyone, take samples of increasing size:

Sample size n	Sample mean weight (example)
5	160 lb
10	154 lb
50	163 lb
100	161 lb
200	162 lb
500	161.5 lb
1,000	161.8 lb

- **Small samples** jump around, because one unusually heavy (230 lb) or light (110 lb) person has a big effect.
- **Large samples** settle down. Here they converge to about **161-162 lb**, which estimates the population mean.

Key takeaways

- More trials or a bigger sample → the observed average gets closer to the true average.
- This is why large samples give reliable estimates of population **parameters**.
- The law says nothing about any single trial. Even after many heads in a row, the next flip is still 50/50.